

Engineering Graphics & Computer Aided Drafting

MEC136 · Study Guide & AutoCAD Manual

Scales ▪ Orthographic Projections ▪ Isometric Views ▪ Surfaces ▪ AutoCAD

Subject Features:

- Complete authoritative coverage of all units in LPU MEC136 Engineering Graphics.
- Standards-compliant drafting principles: BIS SP46, line conventions, dimensioning, and scales.
- Projections of points, lines, planes, and solids using first and third angle methods.
- Isometric transformations, sectional views, surface development, and full AutoCAD command manual.

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How to Use This Book for Exam Success

Welcome to **MEC136: Engineering Graphics & Computer Aided Drafting**. This textbook has been authored from the ground up to serve as the definitive, university-level curriculum companion, laboratory drafting manual, and high-yield exam preparation guide.

Strategic Exam Preparation Paradigms

Scenario A: Full Semester Study Strategy (Maximum Marks + Mastery)

1. **Understand Core Concepts:** Read the topic-by-topic theory, focusing on physical principles (e.g., Similar Triangles in Diagonal Scales, Rotating Line Method for Lines, Glass Box concept for Orthographic Projections).
2. **Master Manual Drafting Procedures:** Follow step-by-step geometric construction steps for line projections, sectional views, and surface developments.
3. **Learn AutoCAD Workflows:** Memorize command shortcuts, options, and key toggles (UNITS, LIMITS, OSNAP F3, ORTHO F8, HATCH, EXTRUDE).
4. **Solve Graded Exercises:** Work through Level 1 to Level 4 practice problems at the end of each unit.

Scenario B: Limited Time / Mid-Term Strategy (High-Yield Focus)

1. **Focus on [HIGH PRIORITY] Topics:** First Angle vs. Third Angle view placement, Line Projections inclined to both planes, Isometric Scale factor derivation (0.816), Sectional hatching exceptions (ribs/fasteners).
2. **Review Key Formulas & Derivations:** Representative Fraction ($R.F.$), Isometric Scale derivation, Sector Angle ($\theta = \frac{r}{L} \times 360^\circ$).
3. **Solve Exam Problems:** Practice 5 solved numerical/construction problems per unit.

Scenario C: Last-Day Night Before Exam Strategy (Rapid Revision)

1. Read **Chapter 7: Master Revision Sheet** (Formula summary, Definition index, AutoCAD Cheat Sheet).
2. Review **Common Student Pitfalls** boxes in Chapters 1–6.
3. Attempt **Chapter 9: Mock Examination** under timed conditions.

Syllabus & Module Correlation Matrix

Unit	Core Module Subject	Chapter	Key Highlights
Unit 1	Introduction to Engineering Drawing	Chapter 1	Instruments, Line Types, Dimensioning, Lettering, Scales (<i>R.F.</i>), AutoCAD Setup.
Unit 2	Projection of Points & Lines	Chapter 2	Quadrants, 1 st vs 3 rd Angle, Point Projections, Line Projections, Traces (<i>HT/VT</i>).
Unit 3	Orthographic Projections of Objects	Chapter 3	Multi-view Projections, Glass Box, Miter Line (45°), Layer & Modify Commands.
Unit 4	Sectional Views & Hatching	Chapter 4	Full/Half/Offset Sections, Hatching Rules (<i>ANSI31</i>), Rib/Fastener Exceptions.
Unit 5	Isometric Views & 3D Modeling	Chapter 5	Isometric Axes (120°/30°), Scale Factor (0.816), Box Method, 3D Primitives.
Unit 6	Surface Development & 3D Editing	Chapter 6	Parallel Line & Radial Line Methods, Truncated Solids, 3D Booleans, VIEWBASE .
Revision	Master Formulas & Commands	Chapter 7	Unit-wise Formula Summary, Derivation Steps, AutoCAD Command Cheat Sheet.
Practice	Cross-Unit Synthesis Problems	Chapter 8	15 Complex Integrated Construction & Calculation Problems with Full Solutions.
Exams	University Assessment Papers	Chapter 9	Two Full 100-Mark Mock Exams (Sections A, B, C) with Marking Rubrics & Solutions.

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Unit 1: Introduction to Engineering Drawing & AutoCAD Setup

1.1 A. What You'll Learn

Engineering drawing is the universal graphic language used by engineers and architects to communicate physical object designs with mathematical precision. In this unit, you will master:

- Essential manual drawing instruments, pencil lead grading, and proper handling.
 - Standardized Line Types (The Alphabet of Lines) defined by ISO and ANSI standards.
 - Dimensioning components, mechanics, Aligned vs. Unidirectional systems, and single-stroke vertical Gothic lettering.
 - Scales, Representative Fraction (*R.F.*), Length of Scale (*L.O.S.*), Plain Scales, and Diagonal Scales based on similar triangles.
 - Fundamental AutoCAD 2D interface navigation, drawing limits, units, object snap (*OSNAP*), orthogonal mode (*ORTHO*), and function keys.
-

1.2 B. Conceptual Framework of Drawing Instruments

1.2.1 Key Instruments and Their Applications

Definition: Drawing Board

Provides a smooth, flat, rectangular surface for securing drawing sheets. Made of seasoned soft wood strips (such as pine) to prevent warping and allow easy pin/tape insertion.

Definition: Mini-Drafter

An indispensable drafting tool that combines the functions of a T-square, set squares, protractor, and scales into a single mechanism. It uses a 3D parallelogram linkage clamped to the drawing board to keep ruling edges parallel across the entire sheet.

- **T-Square:** Traditional tool consisting of a stock and blade used primarily for drawing horizontal parallel lines and supporting set squares.
- **Set Squares:** Used in combination with a drafter or T-square:
 - **45° Set Square:** Angles of 45°, 90°, 135°.
 - **30°–60° Set Square:** Angles of 30°, 60°, 90°, 120°, 150°.

- **Instrument Box:**

- **Large Compass:** Features a knee joint for drawing large circles (> 50 mm radius).
- **Bow Compass:** Used for drawing precise small circles and arcs.
- **Dividers:** Used to transfer dimensions from scales to drawing paper or divide lines into equal segments.

1.2.2 Pencil Lead Grading and Selection

Pencil leads are classified by hardness (clay content vs. graphite ratio).

Important Rule: Pencil Lead Application Matrix

- **Hard Grades (4H to 6H):** High clay content. Used for very light construction lines, center lines, locus lines, and guide lines.
- **Medium Grades (H, 2H):** Standard drafting leads. Used for visible object lines, dimension lines, extension lines, arrowheads, and lettering.
- **Soft Grades (HB, B):** High graphite content. Used for artistic sketching or freehand boundary lines. (Rarely used in precision engineering drawing).

1.2.3 French Curves

French curves are plastic templates with varying smooth curves used to draw non-circular curves (such as ellipses, parabolas, hyperbolas, and involutes) that cannot be drawn with a standard compass.

1.3 C. Line Types (The Alphabet of Lines)

Standardization of lines is governed by ISO and ANSI standards. Line thickness and style convey specific geometric meanings.

Line Type	Representation	Description & Application	Pencil Grade
Visible / Object Line		Continuous thick line. Represents visible outlines and sharp edges.	H or HB
Hidden Line		Dashed medium line. Represents interior/hidden edges.	2H
Center Line		Long dash – short dash chain line. Indicates axes of symmetry.	4H
Construction Line		Continuous very thin line. Used for initial layout projections.	4H or 6H
Dimension / Extension		Continuous thin line. Defines start and limit of measurements.	2H
Cutting Plane Line		Thick chain line with thickened ends. Shows hypothetical section cut.	H

Table 1.1. The Alphabet of Lines: Standard Engineering Line Types.

1.4 D. Dimensioning Principles & Systems

Dimensioning is the process of specifying exact physical sizes, locations, and geometric characteristics of object features.

1.4.1 Components of a Dimension

- **Extension Line:** Thin continuous line extending from the object edge. A gap of 1 mm is left between object and extension line.
- **Dimension Line:** Thin continuous line terminated by arrowheads defining measurement boundaries.
- **Arrowheads:** Standard ratio of **Length to Width** is **3 : 1**. Arrowheads must be sharp, filled, and uniform.
- **Leader Line:** Thin line leading from a note to a feature (e.g., hole diameter), terminating in an arrowhead or dot.

1.4.2 Systems of Dimensioning

Definition: Aligned System

Dimensions are placed perpendicular to the dimension line. Values are placed above the dimension line and read from the bottom or right-hand side of the sheet.

Definition: Unidirectional System

Dimensions are placed vertically regardless of line orientation. Values are read from the bottom of the sheet only. The dimension line is broken in the middle to insert text.

Feature	Aligned Dimensioning System	Unidirectional Dimensioning System
Text Orientation	Perpendicular to dimension line	Always horizontal (upright)
Reading Direction	Read from bottom or right side	Read from bottom only
Dimension Line	Continuous unbroken line	Broken in middle for text entry
Application	Standard engineering drawings (ISO)	Aircraft, automotive, heavy machinery (ANSI)

Table 1.2. Comparison of Aligned and Unidirectional Dimensioning Systems.

1.5 E. Single-Stroke Vertical Gothic Lettering

Lettering on technical drawings must be uniform, highly legible, and suitable for microfilming or digital scanning. "Single stroke" means letter line thickness is produced in one stroke of the pencil.

Single-Stroke Lettering Proportions

- **Standard Height-to-Width Ratio:** 7 : 4 for most letters (*A, B, C, D, E, F, G, H, J, K, L, N, O, P, Q, R, S, T, U, V, X, Y, Z*).
- **Wide Letters (*M, W*):** Ratio of 7 : 5 or 7 : 6.
- **Narrow Letter (*I*):** Single vertical stroke (7 : 1).
- **Letter Spacing:** $\approx \frac{1}{5}h$ (where h is character height).
- **Word Spacing:** $\approx 1h$.
- **Line Spacing:** $\approx \frac{1}{2}h$ to $1h$.

1.6 F. Scales and Representative Fraction (*R.F.*)

Scales allow large objects (e.g., buildings, bridges) or small objects (e.g., watch gears, micro-chips) to be represented accurately on standard paper sizes.

Representative Fraction (*R.F.*)

$$R.F. = \frac{\text{Length of object on drawing sheet}}{\text{Actual physical length of object}} \quad (1.1)$$

Length of Scale (*L.O.S.*)

$$L.O.S. = R.F. \times \text{Maximum length to be measured} \quad (1.2)$$

1.6.1 Classification of Scales

- **Full Scale (1 : 1):** Drawing dimensions equal actual physical dimensions.
- **Reducing Scale (1 : X):** Drawing dimensions smaller than actual object (e.g., 1 : 10, 1 : 50, 1 : 100).
- **Enlarging Scale (X : 1):** Drawing dimensions larger than actual object (e.g., 2 : 1, 5 : 1, 10 : 1).

1.6.2 Plain Scale vs. Diagonal Scale

Definition: Plain Scale

Consists of a line divided into a number of equal main units, with the first main unit subdivided into smaller parts. Represents two units (e.g., Kilometers and Hectometers, or Meters and Decimeters).

Definition: Diagonal Scale

Used when minute measurements are required or to measure three consecutive units (e.g., Meters, Decimeters, and Centimeters) or up to two decimal places. Based on the principle of **Similar Triangles**.

Important Rule: Principle of Similar Triangles in Diagonal Scales

To divide a short vertical line of length h into 10 equal parts, a diagonal line is drawn across 10 horizontal parallel lines. By similar triangles, the distance from the vertical axis to the diagonal line at the n^{th} horizontal line equals $\frac{n}{10}$ of the base unit.

Worked Example: Diagonal Scale Construction

Problem: Construct a diagonal scale of $R.F. = 1 : 40$ to read meters, decimeters, and centimeters up to 6 m. Mark a distance of 4.56 m on the scale.

Solution Step 1: Calculate Length of Scale (L.O.S.)

$$L.O.S. = R.F. \times \text{Max Length} = \frac{1}{40} \times 6 \text{ m} = \frac{1}{40} \times 600 \text{ cm} = \mathbf{15 \text{ cm}}$$

Solution Step 2: Main Division Draw a line 15 cm long and divide it into 6 equal main parts. Each main part represents 1 m.

Solution Step 3: Subdivisions & Diagonals Divide the first left division (1 m) horizontally into 10 equal parts (each = 1 dm = 0.1 m). Erect a vertical rectangle height 5 cm and divide vertically into 10 equal parts (each = 1 cm = 0.01 m). Draw diagonal lines connecting subdivision n on bottom to $n + 1$ on top.

Solution Step 4: Locate 4.56 m Take 4 main units (4 m) to the right of zero, move 5 subdivisions left (0.5 m), and move up to the 6th horizontal line (0.06 m). Total distance = $4 + 0.5 + 0.06 = \mathbf{4.56 \text{ m}}$.

1.7 G. Introduction to AutoCAD Interface & Setup

AutoCAD (Computer-Aided Design) replaces manual drafting tools with digital precision.

1.7.1 Interface Layout Components

1. **Application Menu (Big Red A):** File management (New, Open, Save, Export).
2. **Ribbon:** Organized tabs (Home, Insert, Annotate) containing panels of drafting tools.
3. **Drawing Area (Viewport):** Infinite 2D/3D graphics area.
4. **UCS Icon:** Visual indicator of X, Y, Z coordinate axes.
5. **Command Line:** Primary method for entering AutoCAD commands and viewing system prompts.
6. **Status Bar:** Toggles for drawing aids (Grid, Snap, Ortho, OSNAP) at bottom right.

1.7.2 Essential AutoCAD Setup Commands

COMMAND: UNITS

Defines measurement type and precision.

- **Command:** UNITS or UN
- **Type:** Decimal (Mechanical), Architectural (Feet/Inches).
- **Precision:** Number of decimal places (e.g., 0.00).
- **Insertion Scale:** Units for blocks/content (Millimeters).

COMMAND: LIMITS

Sets the invisible boundary of the drawing area (equivalent to paper size).

- **Command:** LIMITS
- **Lower-Left Corner:** Type 0,0 → Press Enter.
- **Upper-Right Corner:** Type 210,297 (for A4 sheet) → Press Enter.
- **Required Refresh:** Type Z (Zoom) → Enter → Type A (All) → Enter.

1.7.3 Function Key Cheat Sheet

Key	Function	Operational Description
F1	Help	Opens Autodesk help documentation.
F2	Text Screen	Displays command history window.
F3	OSNAP	Toggles Object Snap on/off (Endpoint, Midpoint, Center).
F7	Grid	Toggles background grid display.
F8	ORTHO	Restricts cursor motion to horizontal (0°, 180°) and vertical (90°, 270°).
F9	Snap Mode	Restricts cursor movement to grid intervals.
F10	Polar Tracking	Restricts cursor to specified angle increments.
F12	Dynamic Input	Displays command prompts and coordinates near cursor.

Table 1.3. AutoCAD Function Key Reference.

1.8 H. Chapter 1 Practice Problems & MCQs

1.8.1 Practice Questions

Viva Q&A: Question 1 (Basic)

State the standard height-to-width ratio for single-stroke vertical Gothic lettering for normal letters, wide letters (M, W), and narrow letter (I).

Answer: Normal letters = 7 : 4; Wide letters (M, W) = 7 : 5 or 7 : 6; Narrow letter (I) = 7 : 1.

Viva Q&A: Question 2 (Moderate)

Differentiate between a Plain Scale and a Diagonal Scale. State the underlying geometric principle of a Diagonal Scale.

Answer: A Plain Scale reads two units or one unit and its first subdivision. A Diagonal Scale reads three consecutive units or up to two decimal places. Its underlying principle is the ****Principle of Similar Triangles****.

Viva Q&A: Question 3 (Exam Level)

A distance of 5 cm on a map represents an actual distance of 25 m. Calculate the *R.F.* and determine the Length of Scale (*L.O.S.*) to measure up to 80 m.

Answer:

$$R.F. = \frac{5 \text{ cm}}{25 \text{ m}} = \frac{5}{2500} = \frac{1}{500}$$

$$L.O.S. = R.F. \times \text{Max Length} = \frac{1}{500} \times 8000 \text{ cm} = \mathbf{16 \text{ cm}}$$

1.8.2 Multiple-Choice Questions (MCQs)

1. Which pencil lead grade is recommended for drawing visible object lines? A. 4H B. 6H C. H or 2H D. HB or B **Answer: C** — Medium grades (H, 2H) are used for object lines and dimensioning.

2. What is the standard ratio of length to width for dimension arrowheads? A. 1 : 1 B. 2 : 1 C. 3 : 1 D. 4 : 1 **Answer: C** — Engineering arrowheads require a 3 : 1 length-to-width ratio.

3. The scale factor for a reducing scale 1 : 50 means: A. Drawing is 50 times larger than actual object B. 1 cm on drawing represents 50 cm actual length C. 50 cm on drawing represents 1 cm actual length D. Representative fraction is 50 **Answer: B** — *R.F.* = 1/50, drawing is 1/50th of actual size.

4. Which function key toggles Ortho mode in AutoCAD? A. F3 B. F7 C. F8 D. F12 **Answer: C** — F8 toggles Ortho mode.

5. What is the correct sequence of AutoCAD commands to set up an A4 sheet limit? A. UNITS → LIMITS (0,0 to 210,297) → ZOOM → ALL B. LIMITS → OSNAP → GRID C. LINE → PLINE → RECTANG D. DIMSTYLE → UNITS → SAVE **Answer: A** — Standard setup workflow.

1.9 I. Unit 1 Quick Revision Checklist**Unit Quick Revision Checklist**

- Hard Pencils (4H–6H) → Construction/Center Lines; Medium Pencils (H, 2H) → Object Lines/Lettering.
- Visible Lines → Continuous Thick; Hidden Lines → Dashed Medium; Center Lines → Chain Thin.
- Dimension Arrowhead Ratio = 3 : 1.

- Aligned System → Text perpendicular to dimension line; Unidirectional System → Text horizontal.
- $R.F. = \frac{\text{Drawing Length}}{\text{Actual Length}}$; $L.O.S. = R.F. \times \text{Max Length}$.
- Diagonal Scale Principle = Similar Triangles.
- AutoCAD Setup → UNITS → LIMITS (0,0 to 210,297) → ZOOM ALL.
- F3 = OSNAP, F8 = ORTHO, F7 = GRID.

Unit 2: Projection of Points & Lines

2.1 A. What You'll Learn

Projection is the representation of a 3D physical object onto a 2D plane surface. In this unit, you will master:

- Principal Planes of Projection (*VP, HP, PP*) and the *XY* Reference Line.
 - Principles of Quadrants and comparison of **First Angle Projection** vs. **Third Angle Projection**.
 - Standard point notation and rules for projecting points across all 4 quadrants.
 - Projection of straight lines across 6 fundamental orientation cases.
 - Finding True Length (*TL*) and true inclinations (θ, ϕ) using the **Rotating Line Method**.
 - Locating Horizontal Trace (*HT*) and Vertical Trace (*VT*) of a line.
 - AutoCAD commands for point and line drawing (*LINE, CIRCLE, ARC, PLINE, DIMALIGNED*).
-

2.2 B. Introduction to Orthographic Projections

Definition: Orthographic Projection

A technical drawing method in which parallel projectors extending from an observer at infinity intersect the plane of projection at right angles (90°), forming a 2D image of a 3D object.

2.2.1 Principal Planes and Reference Line

- **Vertical Plane (VP):** Placed vertically in front of the observer. The view obtained on the VP is called the **Front View (FV)** or **Elevation**.
- **Horizontal Plane (HP):** Placed horizontally. The view obtained on the HP is called the **Top View (TV)** or **Plan**.
- **Reference Line (XY Line):** The line of intersection between the HP and VP.
- **Profile Plane (PP):** Auxiliary plane placed perpendicular to both HP and VP, used for **Side Views**.

2.2.2 Principles of Quadrants and Projection Systems

The intersection of VP and HP creates four quadrants (1st, 2nd, 3rd, 4th).

Feature	First Angle Projection (ISO)	Third Angle Projection (ANSI)
Object Position	1 st Quadrant (Above HP, In front of VP)	3 rd Quadrant (Below HP, Behind VP)
Observer Logic	Observer → Object → Plane	Observer → Plane → Object
View Position	Front View is above XY; Top View is below XY.	Top View is above XY; Front View is below XY.
Side View	Left Side View drawn to right of FV.	Right Side View drawn to right of FV.
Usage	Standard in India, Europe, Asia (ISO)	Standard in USA, Canada (ANSI)

Table 2.1. Comparison of First Angle and Third Angle Projection Systems.

2.3 C. Orthographic Projection of Points

A point has position in space but no length, width, or thickness.

Important Rule: Standard Point Notation

- **Actual Point in Space:** Capital Letter (A, B, C).
- **Front View (on VP):** Lowercase letter with a dash (a', b', c').
- **Top View (on HP):** Lowercase letter without a dash (a, b, c).
- **Side View (on PP):** Lowercase letter with double dash (a'', b'', c'').

2.3.1 Rules for Projection of Points Across 4 Quadrants

Quadrant	Spatial Position	Front View (a')	Top View (a)
1 st	Above HP, In front of VP	Above XY line	Below XY line
2 nd	Above HP, Behind VP	Above XY line	Above XY line (Overlapping)
3 rd	Below HP, Behind VP	Below XY line	Above XY line
4 th	Below HP, In front of VP	Below XY line	Below XY line (Overlapping)

Table 2.2. Point Projection Rules across Quadrants.

Worked Example: Point Projection Problem

Problem: Draw the projections of the following points on a common reference line XY : 1. Point A is 20 mm above HP and 30 mm in front of VP. 2. Point B is 25 mm below HP and 40 mm behind VP. 3. Point C is in the HP and 35 mm behind VP.

Solution:

1. **Point A (1st Quadrant):** Draw projector line perpendicular to XY . Mark a' 20 mm above XY and a 30 mm below XY .
2. ****Point B (3rd Quadrant):**** Mark b' 25 mm below XY and b 40 mm above XY .
3. ****Point C (In HP, Behind VP):**** c' lies on XY line; c is 35 mm above XY .

2.4 D. Projection of Straight Lines

A line is the shortest path connecting two points. Projection of a line depends on its orientation relative to HP and VP.

2.4.1 Fundamental Orientation Cases

1. Line Parallel to Both HP and VP:

- Front View: Horizontal line parallel to XY showing True Length (TL).
- Top View: Horizontal line parallel to XY showing True Length (TL).

2. Line Perpendicular to HP and Parallel to VP:

- Front View: Vertical line perpendicular to XY showing True Length (TL).
- Top View: A point ($a \equiv b$).

3. Line Perpendicular to VP and Parallel to HP:

- Front View: A point ($a' \equiv b'$).
- Top View: Vertical line perpendicular to XY showing True Length (TL).

4. Line Inclined to HP (θ) and Parallel to VP:

- Front View: Inclined line showing True Length (TL) and true angle θ .
- Top View: Horizontal line parallel to XY showing apparent shorter length ($ab = TL \cos \theta$).

5. Line Inclined to VP (ϕ) and Parallel to HP:

- Front View: Horizontal line parallel to XY showing apparent shorter length ($a'b' = TL \cos \phi$).
- Top View: Inclined line showing True Length (TL) and true angle ϕ .

6. Line Inclined to Both HP (θ) and VP (ϕ):

- Neither view shows True Length directly.
- Front View shows apparent length and apparent inclination angle α .
- Top View shows apparent length and apparent inclination angle β .
- Requires the **Rotating Line Method**.

2.5 E. The Rotating Line Method for Lines Inclined to Both Planes

Governing Relationships for Oblique Lines

$$\text{Front View Apparent Length } (a'b') = TL \cos \phi \quad (2.1)$$

$$\text{Top View Apparent Length } (ab) = TL \cos \theta \quad (2.2)$$

$$\text{Height Difference } (\Delta h) = TL \sin \theta, \quad \text{Depth Difference } (\Delta d) = TL \sin \phi$$

2.5.1 Step-by-Step Construction Procedure

1. **Reference Line & End A:** Draw XY line. Plot locus of A : front view a' above XY and top view a below XY .
2. ****True Length Inclination Lines:****
 - From a' , draw line $a'b'_1$ equal to TL at true angle θ to horizontal.
 - From a , draw line ab_2 equal to TL at true angle ϕ to horizontal.
3. ****Locus Lines:**** Draw horizontal locus lines through b'_1 (Locus of B in FV) and through b_2 (Locus of B in TV).
4. ****Apparent Front View Construction:****
 - Project b_2 vertically up to meet horizontal line from a' at point b'_2 .
 - With center a' and radius $a'b'_2$, draw an arc to intersect locus of B in FV at point b' .
 - Join $a'b'$ to get final ****Front View**** (apparent angle α).
5. ****Apparent Top View Construction:****
 - Project b'_1 vertically down to meet horizontal line from a at point b_1 .
 - With center a and radius ab_1 , draw an arc to intersect locus of B in TV at point b .
 - Join ab to get final ****Top View**** (apparent angle β).
6. ****Check:**** Points b' and b MUST lie on the same vertical projector line!

Worked Example: Line Inclined to Both Planes

Problem: A line AB , 75 mm long, has its end A 15 mm above HP and 20 mm in front of VP. The line is inclined at 45° to HP and 30° to VP. Draw its projections.

Solution Step 1: Given Data

$TL = 75$ mm, End $A : a' = 15$ mm above XY , $a = 20$ mm below XY , $\theta = 45^\circ$, $\phi = 30^\circ$

Solution Step 2: Calculate Apparent Lengths

Top View Length (ab_1) = $TL \cos \theta = 75 \cos 45^\circ = 75 \times 0.707 = \mathbf{53.03}$ mm

Front View Length ($a'b'_2$) = $TL \cos \phi = 75 \cos 30^\circ = 75 \times 0.866 = \mathbf{64.95}$ mm

Solution Step 3: Draw Projections Follow 6-step Rotating Line construction procedure to plot $a'b'$ and ab .

2.6 F. Concept of Traces

Traces are the points where a line (extended if necessary) intersects the reference planes.

Definition: Horizontal Trace (HT)

The point of intersection of the line with the Horizontal Plane (HP).

- To locate HT: Extend Front View $a'b'$ to meet XY at point h . From h , erect a

perpendicular to intersect the extended Top View ab at point HT.

Definition: Vertical Trace (VT)

The point of intersection of the line with the Vertical Plane (VP).

- To locate VT: Extend Top View ab to meet XY at point v . From v , erect a perpendicular to intersect the extended Front View $a'b'$ at point VT.

2.7 G. AutoCAD Commands for Points and Lines

COMMAND: LINE

Creates straight line segments.

- **Command:** LINE or L
- **Absolute Coordinates:** Type L → Enter → Type X,Y (e.g., 10,20).
- **Polar Coordinates:** Type @length<angle (e.g., @75<45).

COMMAND: PLINE

Creates 2D polylines (connected sequence of line/arc segments forming a single object).

- **Command:** PLINE or PL
- **Difference from Line:** A rectangle drawn with LINE consists of 4 separate objects; drawn with PLINE it is 1 unified object.

COMMAND: DIMALIGNED

Measures true length of inclined lines.

- **Command:** DIMALIGNED or DAL
- **Usage:** Click start endpoint → Click end endpoint → Drag dimension position.

2.8 H. Chapter 2 Practice Problems & MCQs

2.8.1 Practice Questions

Viva Q&A: Question 1 (Basic)

If a point lies in the 3rd quadrant, state the position of its Front View and Top View relative to the XY line.

Answer: Front View (a') is **below** XY line; Top View (a) is **above** XY line.

Viva Q&A: Question 2 (Moderate)

What is a Horizontal Trace (HT)? Describe the procedure to locate HT.

Answer: HT is the point where a line intersects the Horizontal Plane. To locate HT, extend Front View $a'b'$ to intersect XY at h , then project vertically to meet the extended Top View ab at HT.

Viva Q&A: Question 3 (Exam Level)

A line PQ 80 mm long is inclined at 30° to HP and 45° to VP. End P is 20 mm above HP and 25 mm in front of VP. Calculate the apparent lengths of Front View and Top View.

Answer:

$$\text{Front View Apparent Length} = TL \cos \phi = 80 \cos 45^\circ = 80 \times 0.707 = \mathbf{56.57 \text{ mm}}$$

$$\text{Top View Apparent Length} = TL \cos \theta = 80 \cos 30^\circ = 80 \times 0.866 = \mathbf{69.28 \text{ mm}}$$

2.8.2 Multiple-Choice Questions (MCQs)

1. In First Angle Projection, where is the Top View drawn? A. Above Front View B. Below Front View C. To the left of Front View D. On the XY line **Answer: B** — Top View is drawn below Front View in First Angle Projection.
2. If a line is parallel to VP and perpendicular to HP, its Top View will be a: A. Line showing True Length B. Line parallel to XY C. Point D. Inclined line **Answer: C** — Top view of a line perpendicular to HP is a point.
3. The point where the extended front view of a line meets the XY reference line is called: A. Horizontal Trace (HT) B. Vertical Trace (VT) C. Point h D. Point v **Answer: C** — Meeting point of FV on XY is h ; projecting vertically to TV gives HT.
4. Which AutoCAD command measures the exact length of an inclined line? A. DIMLINEAR B. DIMALIGNED C. DIMANGULAR D. DIST **Answer: B** — DIMALIGNED measures true length of inclined lines.
5. What is the relation between true length (TL) and apparent front view length (FVL) when a line is inclined to VP at angle ϕ ? A. $FVL = TL \sin \phi$ B. $FVL = TL \cos \phi$ C. $FVL = TL \tan \phi$ D. $FVL = TL$ **Answer: B** — $FVL = TL \cos \phi$.

2.9 I. Unit 2 Quick Revision Checklist

Unit Quick Revision Checklist

- 1st Angle: FV above XY , TV below XY ; 3rd Angle: TV above XY , FV below XY .
- Point Notation: Capital A = Space, a' = FV (on VP), a = TV (on HP).
- Line $\parallel$ to plane $\rightarrow$ Shows True Length TL ; Line $\perp$ to plane $\rightarrow$ Appears as a Point.
- Front View Apparent Length = $TL \cos \phi$; Top View Apparent Length = $TL \cos \theta$.
- Rotating Line Method: Projectors b' and b must align vertically.
- HT location: Extend FV $\rightarrow h$ on $XY \rightarrow$ project to extended TV $\rightarrow$ HT.
- VT location: Extend TV $\rightarrow v$ on $XY \rightarrow$ project to extended FV $\rightarrow$ VT.
- AutoCAD: LINE (@length<angle), DIMALIGNED for inclined lengths.

Unit 3: Orthographic Projections of Objects

3.1 A. What You'll Learn

In this unit, you transition from points and lines to complete 3D mechanical components and objects. You will master:

- Multi-view orthographic projection theory and the **Glass Box Concept**.
 - ISO First Angle vs. ANSI Third Angle view arrangements and symbol differentiation.
 - The **Miter-Line Method (45°)** for depth transfer between Top View and Side Views.
 - Application of visible object lines, hidden lines, and center lines on engineering views.
 - AutoCAD Layer Management ('LINTYPE', 'LTSCALE').
 - AutoCAD 2D Modification commands ('MOVE', 'COPY', 'ROTATE', 'TRIM', 'MIRROR', 'SCALE', 'FILLET', 'CHAMFER', 'ARRAY').
 - Complete step-by-step AutoCAD 2D drafting workflow for multi-view orthographic drawings.
-

3.2 B. Multi-View Orthographic Projection Principles

3.2.1 The "Glass Box" Concept

Imagine a 3D object suspended inside a transparent glass box.

1. The observer looks perpendicular (90°) to each face of the box.
2. Points and features of the object are projected onto the transparent glass walls.
3. The glass box is unfolded flat into a single 2D plane surface.
4. The resulting 2D drawings form the set of orthographic views: **Front View (FV)**, **Top View (TV)**, **Left Side View (LSV)**, and **Right Side View (RSV)**.

3.2.2 View Layout Systems: First Angle vs. Third Angle

Important Rule: First Angle View Placement (ISO Standard)

The object is between Observer and Plane of Projection.

- **Front View (FV):** Placed above the *XY* reference line.
- **Top View (Plan):** Placed directly **below** the Front View.
- **Left Side View (LSV):** Drawn to the **right** of the Front View.

- **Right Side View (RSV):** Drawn to the **left** of the Front View.

Important Rule: Third Angle View Placement (ANSI Standard)

The Plane of Projection is between Observer and Object.

- **Front View (FV):** Placed below the *XY* reference line.
- **Top View (Plan):** Placed directly **above** the Front View.
- **Right Side View (RSV):** Drawn to the **right** of the Front View.
- **Left Side View (LSV):** Drawn to the **left** of the Front View.

3.3 C. The Miter-Line Method for View Projection

When constructing 2D orthographic projections manually or in AutoCAD, dimensions are correlated across views:

- **Front View & Top View:** Share identical **Length (Width)** dimensions horizontally.
- **Front View & Side View:** Share identical **Height** dimensions vertically.
- **Top View & Side View:** Share identical **Depth** dimensions.

Miter Line (45°) Depth Transfer

To project depth dimensions accurately between Top View and Side View:

1. Draw a diagonal construction line at 45° starting from the intersection of the main reference axes in the projection quadrant.
2. Project horizontal lines from Top View features to meet the 45° Miter Line.
3. At each intersection point on the Miter Line, project vertical lines down (or up) into the Side View area.
4. Project horizontal lines from Front View across into the Side View. The grid intersection forms the Side View geometry.

3.4 D. AutoCAD Layer Properties & Line Management

Technical drawings rely on standardized line properties managed via the **Layer Properties Manager**.

COMMAND: LTSCALE

Globally scales the length of dashes and gaps in non-continuous line types (e.g., **DASHED**, **CENTER**).

Layer Name	Color	Line Type	Line Weight & Purpose
Object Lines	White / Black	Continuous	0.5–0.7 mm (Thick visible outlines)
Hidden Lines	Yellow	DASHED	0.3 mm (Medium interior hidden details)
Center Lines	Red	CENTER	0.3 mm (Thin long-short-long axes)
Projectors	Grey / Cyan	Continuous	0.15 mm (Thin layout guide lines)
Dimensions	Green	Continuous	0.25 mm (Dimension text and arrows)

Table 3.1. Standard AutoCAD Layer Property Configuration.

- **Command:** LTSCALE
- **Usage:** If dashed center lines appear as continuous solid lines, type **LTSCALE** → Enter scale factor (e.g., 0.5 or 10).

3.5 E. AutoCAD 2D Modification Commands

COMMAND: MOVE & COPY

Displaces or duplicates objects.

- **Move Command:** MOVE or M
- **Copy Command:** COPY or CO / CP
- **Workflow:** Select objects → Press **Enter** → Specify Base Point → Specify Second Point.

COMMAND: ROTATE

Revolves selected objects around a base point.

- **Command:** ROTATE or RO
- **Direction:** Positive angle values rotate ****Counter-Clockwise****; negative values rotate Clockwise.

COMMAND: TRIM

Cuts objects to meet the edges of other boundary objects.

- **Command:** TRIM or TR
- **Workflow:** Select cutting edges (or press **Enter** for select-all) → Click object segments to remove.

COMMAND: MIRROR

Creates a reverse symmetrical copy of selected objects across a defined axis.

- **Command:** MIRROR or MI
- **Workflow:** Select objects → Specify 1st and 2nd points of mirror line → Choose whether to delete source objects (Yes/No).

COMMAND: FILLET & CHAMFER

Rounds or bevels corner edges of objects.

- **Fillet Command:** FILLET or F (Set radius first: Type F → R → Enter radius value → Select two lines).
- **Chamfer Command:** CHAMFER or CHA (Set distance first: Type CHA → D → Enter Dist 1 & Dist 2 → Select two lines).

COMMAND: ARRAY

Creates multiple copies of objects in a rectangular, polar, or path pattern.

- **Command:** ARRAY or AR
- **Rectangular Array:** Arranges copies into Rows and Columns grid.
- **Polar Array:** Copies objects in a circular pattern around a center axis point.

3.6 F. Hands-on AutoCAD Workflow: 2D Orthographic Views

Follow this exact multi-step procedure when converting a 3D isometric view into 2D orthographic drawings:

1. **Setup Phase:**

- Set units: Type UN → Decimal, 0.00 precision, Millimeters.
- Set limits: Type LIMITS → 0,0 to 210,297 → Type Z → A.
- Create Layers: Object Lines (Thick White), Hidden Lines (Yellow Dashed), Center Lines (Red Center), Projectors (Grey).

2. ****Construction Grid:****

- Draw horizontal and vertical reference lines using XLINE or LINE.
- Enable ORTHO (F8) for straight projection lines.

3. ****Front View Construction:****

- Switch to 'Object Layer'.
- Draw Front View outline using LINE, CIRCLE, FILLET.

4. ****Top View Construction:****

- Switch to 'Projector Layer'. Use XLINE to project vertical lines down from Front View corners.
- Switch to 'Object Layer' and draw Top View width and depth outlines.
- Add hidden lines for internal holes using 'Hidden Layer'.

5. ****Side View Construction (Miter Line Method):****

- Draw a 45° line from reference axis intersection.
- Project horizontal lines from Top View rightwards to meet 45° line.
- Project vertical lines down from 45° line into Side View quadrant.
- Project horizontal lines from Front View into Side View quadrant.
- Trace Side View outlines at grid intersections.

6. Final Details & Dimensioning:

- Mark hole centers using 'Center Layer'.
- Add linear dimensions (DLI) and radial dimensions (DRA).

3.7 G. Chapter 3 Practice Problems & MCQs

3.7.1 Practice Questions

Viva Q&A: Question 1 (Basic)

Explain the "Glass Box Concept" in orthographic projections.

Answer: The 3D object is imagined suspended inside a transparent glass box. Views are projected perpendicularly onto the faces of the box, which is then unfolded flat into a 2D plane.

Viva Q&A: Question 2 (Moderate)

In First Angle Projection, where are the Top View and Left Side View placed relative to the Front View?

Answer: Top View is placed **below** the Front View. Left Side View is drawn to the **right** of the Front View.

Viva Q&A: Question 3 (Exam Level)

Describe the steps to draw a Fillet of radius 15 mm on a sharp corner in AutoCAD.

Answer: Type FILLET (or F) → Press **Enter** → Type R (Radius) → Press **Enter** → Type 15 → Press **Enter** → Click first line → Click second line.

3.7.2 Multiple-Choice Questions (MCQs)

1. In First Angle Projection, the Right Side View is drawn to the: A. Right of Front View B. Left of Front View C. Top of Front View D. Bottom of Front View **Answer: B** — Right Side View is drawn to the left of Front View in First Angle Projection.
2. Which AutoCAD command is used to round sharp corners with a specific radius? A. CHAMFER B. FILLET C. ARC D. CIRCLE **Answer: B** — FILLET rounds sharp corners.
3. What angle is used for the Miter Line to transfer depth between Top View and Side View? A. 30° B. 45° C. 60° D. 90° **Answer: B** — The Miter Line is drawn at 45°.
4. Which line type is used to represent hidden interior edges in orthographic views? A.

Continuous Thick Line B. Long dash – short dash Chain Line C. Dashed Medium Line D. Continuous Thin Line **Answer: C** — Dashed lines represent hidden interior edges.

5. What does the AutoCAD command LTSCALE control? A. Scale of drawing limits B. Global dash and gap spacing of non-continuous lines C. Text height in dimension style D. Zoom magnification level **Answer: B** — LTSCALE controls global non-continuous line dash scale.

3.8 H. Unit 3 Quick Revision Checklist

Unit Quick Revision Checklist

- First Angle: FV above XY, TV below FV, LSV to right of FV, RSV to left of FV.
- Third Angle: FV below XY, TV above FV, RSV to right of FV, LSV to left of FV.
- Depth Transfer: Use 45° Miter Line between TV and Side View.
- Visible Outlines → 0.5–0.7 mm Thick; Hidden Lines → 0.3 mm Dashed; Center Lines → Chain Line.
- AutoCAD Fillet: F → R → Radius Value → Select Lines.
- AutoCAD Chamfer: CHA → D → Dist 1, Dist 2 → Select Lines.
- AutoCAD Rotate: Positive angle rotates Counter-Clockwise.
- AutoCAD Array: Rectangular (Rows/Cols) vs. Polar (Circular).

Unit 4: Sectional Views & Hatching Standards

4.1 A. What You'll Learn

Standard orthographic views represent hidden internal features using dashed lines. However, when an object has complex internal geometry, an abundance of hidden lines creates visual confusion. In this unit, you will master:

- Purpose and principles of sectioning.
 - Cutting Plane Lines and Section Hatching standards (45° thin continuous lines, ANSI31).
 - **Full Section**, **Half Section**, and **Offset Section** view types.
 - Sectioning view placement in First Angle and Third Angle systems.
 - **Non-hatching exceptions** (webs/ribs, fasteners, shafts, pins, solid rods).
 - AutoCAD sectioning and modification tools (STRETCH, EXPLODE, OFFSET, EXTEND, JOIN, REGION, BREAK).
 - AutoCAD Hatching commands (HATCH, HATCHEDIT, ANSI31, Pick Points vs. Select Objects).
-

4.2 B. Fundamentals of Sectioning

Definition: Sectional View

A drawing view created by imagining an object cut by a plane surface to expose its internal construction details.

4.2.1 Purpose of Sectioning

- To clarify internal details and eliminate confusion caused by excessive hidden dashed lines.
- To facilitate accurate dimensioning of internal features (holes, keyways, slots).
- To clearly distinguish solid material from internal void spaces.

4.2.2 Key Components of Sectioning

1. **Cutting Plane Line:** A thick chain line (long dash alternating with two short dashes) with thickened ends and arrowheads indicating the direction of viewing.
 2. **Section Lines (Hatching):** Thin continuous lines drawn at a **45°** angle to indicate solid material cut by the plane. Void areas (holes/cavities) are **NOT** hatched.
-

4.3 C. Types of Sectional Views

4.3.1 1. Full Section

Created when the cutting plane passes entirely through the object in a straight line, splitting it into two parts.

- **Usage:** Best for objects where internal features lie along a central axis.
- **Representation:** The sectional view replaces the standard exterior view.

4.3.2 2. Half Section

Created by passing two cutting planes at right angles to each other along center lines. Effectively, **one-quarter of the object is removed**.

- **Usage:** Ideal for symmetrical objects (cylinders, pulleys, gear blanks).
- **Representation:** One half of the view shows internal features in section; the other half shows exterior appearance.
- **Key Rule:** Hidden lines are omitted in the un-sectioned half. The boundary between sectioned and un-sectioned halves is a **Center Line**, not a solid line!

4.3.3 3. Offset Section

Used when internal features do not lie in a straight line. The cutting plane "steps" or bends to pass through offset features.

- **Usage:** Complex parts with holes/slots located at different depths.
 - **Key Rule:** The bends in the cutting plane are **NOT** shown in the final sectional view; the view appears as if all features lay on a single continuous plane.
-

4.4 D. Non-Hatching Exceptions (Rules of Sectioning)

Important Rule: Standard Engineering Sectioning Exceptions

To prevent misinterpretation of solid strength or shape, the following solid items are **NOT** hatched when cut longitudinally (along their length):

1. **Thin Ribs / Webs:** Triangular or flat reinforcing webs cut along their longitudinal plane are left unhatched.
 2. **Fasteners:** Bolts, nuts, screws, studs, washers, and rivets.
 3. **Shafts & Solid Pins:** Solid shafts, keys, cotters, and pins.
 4. **Ball and Roller Bearings:** Rolling elements (balls, rollers).
-

4.5 E. AutoCAD Modification & Creation Commands

COMMAND: STRETCH

Elongates or shrinks geometry while maintaining vertex connectivity.

- **Command:** STRETCH or S
- **Critical Requirement:** Objects **MUST** be selected using a ****Crossing Window**** (green window drawn right-to-left). Points ***inside*** the window move; points ***outside*** remain fixed.

COMMAND: EXPLODE

Breaks compound objects (Blocks, Polylines, Hatches, Rectangles) into constituent component segments (lines, arcs).

- **Command:** EXPLODE or X

COMMAND: OFFSET

Creates concentric circles, parallel lines, or parallel curves at a specified distance.

- **Command:** OFFSET or O
- **Workflow:** Type O → Enter distance value → Select object → Click side to offset.

COMMAND: EXTEND

Lengthens a line or arc to meet another boundary object edge.

- **Command:** EXTEND or EX

COMMAND: REGION

Converts closed coplanar line segments into a 2D region plane surface.

- **Command:** REGION or REG

4.6 F. AutoCAD Hatching Commands & Workflow

COMMAND: HATCH

Fills enclosed areas or selected objects with a hatch pattern or solid fill.

- **Command:** HATCH or H
- **Standard Pattern:** ANSI31 (diagonal lines at 45° for cast iron / general material).

- **Hatch Scale:** Adjusts line spacing (if hatching appears solid black, increase scale; if lines don't appear, decrease scale).
- **Pick Points vs. Select Objects:**
 - **Pick Points:** Click inside a closed boundary area (AutoCAD auto-detects boundary).
 - **Select Objects:** Click specific closed boundary objects.

COMMAND: HATCHEDIT

Modifies an existing hatch pattern, scale, or angle without redrawing.

- **Command:** HATCHEDIT or HE

4.6.1 Step-by-Step 2D Sectional View Workflow

1. Draw standard Top View and Front View outline. 2. Draw Cutting Plane Line on Top View (e.g., passing through center). 3. Project cut feature boundary lines down into Front View using **XLINE**. 4. Trim away exterior shell lines cut away by plane using **TRIM**. 5. Change exposed internal hidden dashed lines to continuous visible object lines. 6. Switch to 'Hatch Layer' → Type H → Select pattern **ANSI31** → Click inside solid cut walls. (Ensure webs and fasteners are excluded!). 7. Label view below: "SECTION A-A".

4.7 G. Chapter 4 Practice Problems & MCQs

4.7.1 Practice Questions

Viva Q&A: Question 1 (Basic)

What is a Half Section? What line separates the sectioned half from the un-sectioned half?

Answer: A Half Section removes one-quarter of a symmetrical object using two cutting planes at right angles. The boundary line between sectioned and un-sectioned halves is a ****Center Line****.

Viva Q&A: Question 2 (Moderate)

State three solid components that are exempt from hatching when cut longitudinally.

Answer: 1. Thin Reinforcing Webs/Ribs. 2. Fasteners (Bolts, Nuts, Screws). 3. Solid Shafts and Pins.

Viva Q&A: Question 3 (Exam Level)

Describe the AutoCAD procedure to stretch a shaft length by 30 mm using the **STRETCH** command.

Answer: Type **STRETCH** (or **S**) → Press **Enter** → Draw a ****Crossing Window**** (green window right-to-left) enclosing the shaft end → Press **Enter** → Click base point → Move cursor in stretch direction with Ortho ON → Type **30** → Press **Enter**.

4.7.2 Multiple-Choice Questions (MCQs)

1. What is the standard angle for general section hatching lines (ANSI31)? A. 30° B. 45° C. 60° D. 90° **Answer: B** — Section hatching lines are drawn at a 45° angle.
2. Which section type removes one-quarter of an object? A. Full Section B. Half Section C. Offset Section D. Revolved Section **Answer: B** — Half section removes one-quarter of the object.
3. Thin reinforcing webs cut longitudinally are: A. Hatched with double lines B. Hatched at 60° C. Left unhatched D. Drawn with dashed lines **Answer: C** — Webs cut longitudinally are left unhatched.
4. Which selection method **MUST** be used with the AutoCAD **STRETCH** command? A. Single Pick Click B. Blue Window Selection (Left-to-Right) C. Green Crossing Window Selection (Right-to-Left) D. Select All **Answer: C** — **STRETCH** requires a green Crossing Window selection.
5. What does an Offset Section cutting plane line do? A. Passes straight through the central axis B. Steps/bends to pass through offset features C. Cuts at a 45° angle D. Removes half the object **Answer: B** — Offset section cutting planes step/bend to pass through offset features.

4.8 H. Unit 4 Quick Revision Checklist

Unit Quick Revision Checklist

- Section Hatching Line Angle = 45° (ANSI31 Pattern).
- Half Section → 1/4 of object removed; boundary line = Center Line.
- Offset Section → Cutting plane steps/bends; bends NOT shown in section view.
- Non-Hatching Exceptions → Webs/ribs, bolts, nuts, shafts, pins cut longitudinally.
- AutoCAD Stretch → Must use Green Crossing Window (Right-to-Left).
- AutoCAD Hatch → **H** → ANSI31 → Pick Points.
- AutoCAD Explode → **X** breaks compound polylines/hatches into separate lines.

Unit 5: Isometric Views & 3D Modeling Fundamentals

5.1 A. What You'll Learn

Isometric projection is a pictorial drawing method enabling a 3D object to be visualized on a 2D plane surface. In this unit, you will master:

- Isometric principles, Isometric Axes (120° separation, 30° receding axes), Isometric Lines, and Non-Isometric Lines.
- Mathematical derivation of the **Isometric Scale Reduction Factor (0.816)**.
- Differentiation between **Isometric View (Drawing)** (True Scale 1 : 1) and **Isometric Projection** (Reduced Scale 0.816).
- The **Box Method** for constructing prisms, pyramids, and composite/stacked solids.
- Isometric dimensioning rules and obliqued text angles ($-30^\circ / +30^\circ$).
- AutoCAD 3D Modeling Coordinate Systems ('3P UCS'), 3D Solid Primitives ('BOX', 'CYLINDER', 'CONE', 'SPHERE'), Extrude ('EXTRUDE'), Revolve ('REVOLVE'), and Presspull ('PRESSPULL').

5.2 B. Isometric Principles & Terminology

Definition: Isometric Projection

A type of axonometric pictorial projection where three mutually perpendicular edges of a cube are equally inclined to the plane of projection. Consequently, all three axes are foreshortened by the exact same amount.

5.2.1 Key Isometric Characteristics

- **Isometric Axes:** Three principal lines meeting at a point and making an angle of **120°** with each other.
 - Vertical Axis: Represents height.
 - Right Axis (30° to horizontal): Represents length/width.
 - Left Axis (30° to horizontal): Represents width/length.
- **Isometric Lines:** Any line parallel to an isometric axis. Dimensions along isometric lines can be measured directly.
- **Non-Isometric Lines:** Any line NOT parallel to an isometric axis (e.g., slant edges of

pyramids, diagonals of cubes). ****Non-isometric lines cannot be measured directly****; their endpoints must be located using isometric coordinates.

- **Isometric Planes:** The three faces of a cube formed by pairs of isometric axes (Top plane, Left vertical plane, Right vertical plane).

5.3 C. Isometric Scale Derivation & Scale Factor

In a true isometric projection, edges are foreshortened because the cube is tilted relative to the projection plane.

Mathematical Derivation of Isometric Scale Factor

Consider a cube top face tilted at 45° to the true horizontal line and 30° to the isometric axis line:

$$\frac{\text{Isometric Length}}{\text{True Length}} = \frac{\cos 45^\circ}{\cos 30^\circ} = \frac{1/\sqrt{2}}{\sqrt{3}/2} = \sqrt{\frac{2}{3}} \approx \mathbf{0.8165} \quad (5.1)$$

$$\text{Isometric Length} = \mathbf{0.816} \times \text{True Length} \quad (5.2)$$

5.3.1 Isometric View vs. Isometric Projection

Parameter	Isometric View (Drawing)	Isometric Projection
Scale Used	**True Scale** (1 : 1, actual physical dimensions)	**Isometric Scale** (0.816 × True Length)
Size	Larger ($\approx 22.5\%$ larger than projection)	Real foreshortened size
Constructibility	Fast and simple to construct directly	Requires constructing an isometric scale line
Standard Usage	Standard practice in AutoCAD and CAD drafting	Theoretical descriptive geometry examinations

Table 5.1. Comparison of Isometric View and Isometric Projection.

5.4 D. Construction Methods for Isometric Solids

5.4.1 The Box Method (Enclosing Box Method)

Solids are constructed by imagining the object enclosed inside a rectangular box of overall Length (L), Width (W), and Height (H).

Procedure for Prisms

1. Draw the three isometric axes meeting at origin (120° apart, two receding at 30°).
2. Construct the base polygon on the isometric plane inside an enclosing rectangle.
3. Erect vertical isometric lines from all base corners equal to height H .
4. Connect top points to form top face.

Procedure for Pyramids

1. Enclose base polygon in a rectangle on the bottom isometric plane. 2. Draw diagonals across the base to locate center point O . 3. From O , erect a vertical line equal to axis height H to locate Apex P . 4. Connect Apex P to base corners. (Note: Slant edges are **non-isometric lines**).

5.4.2 Composite / Stacked Solids

For objects stacked on top of one another (e.g., a cylinder on a cube): 1. Construct the bottom solid completely. 2. Locate center point of top face of bottom solid using diagonals. 3. Construct base of upper solid centered on that point. 4. Draw upper solid and erase hidden obscured lines.

5.5 E. AutoCAD 3D Modeling Commands

Transitioning from 2D drafting to 3D solid modeling requires mastering coordinate systems and 3D primitives.

COMMAND: 3P UCS (3-Point User Coordinate System)

Moves the coordinate system working plane to any face of a 3D object.

- **Command:** UCS → Type 3 (3 Point)
- **Workflow:** Specify new Origin (0, 0, 0) → Specify point on positive X-axis → Specify point on positive Y-axis.

5.5.1 3D Solid Primitives

- **BOX:** Creates a rectangular solid prism (L, W, H).
- **CYLINDER:** Creates a cylinder with defined base radius and height.
- **CONE:** Creates a cone with defined base radius and height.
- **SPHERE:** Creates a solid ball with defined radius.

COMMAND: EXTRUDE

Converts closed 2D polylines, circles, or regions into 3D solids by stretching along Z-axis.

- **Command:** EXTRUDE or EXT
- **Input:** Closed polyline, circle, or region.
- **Workflow:** Select object → Enter height → Enter taper angle (optional).

COMMAND: REVOLVE

Creates a solid by sweeping a 2D closed profile around an axis line.

- **Command:** REVOLVE or REV
- **Workflow:** Select 2D profile → Select axis line → Enter angle of revolution (usually 360°).

COMMAND: PRESSPULL

Dynamically detects enclosed coplanar boundary areas and adds/subtracts 3D volume.

- **Command:** PRESSPULL
- **Advantage over Extrude:** Does not require a polyline; automatically detects enclosed areas bounded by coplanar lines.

5.6 F. Chapter 5 Practice Problems & MCQs

5.6.1 Practice Questions

Viva Q&A: Question 1 (Basic)

What are non-isometric lines? Can dimensions be measured directly along them?

Answer: Non-isometric lines are lines that are NOT parallel to any of the three isometric axes (e.g., slant edges of a pyramid). Dimensions **cannot** be measured directly along them; endpoints must be plotted using isometric coordinates.

Viva Q&A: Question 2 (Moderate)

Derive the isometric scale factor (0.816).

Answer: Ratio of isometric length to true length = $\frac{\cos 45^\circ}{\cos 30^\circ} = \frac{1/\sqrt{2}}{\sqrt{3}/2} = \sqrt{\frac{2}{3}} \approx \mathbf{0.816}$.

Viva Q&A: Question 3 (Exam Level)

Calculate the isometric length corresponding to a true length of 90 mm.

Answer:

$$\text{Isometric Length} = 0.816 \times \text{True Length} = 0.816 \times 90 \text{ mm} = \mathbf{73.44 \text{ mm}}$$

5.6.2 Multiple-Choice Questions (MCQs)

1. What is the angle between any two principal isometric axes? A. 30° B. 60° C. 90° D. 120°

Answer: D — Isometric axes meet at 120° intervals.

2. The ratio of isometric length to true length is approximately: A. 0.577 B. 0.707 C. 0.816 D. 0.866 **Answer: C** — The ratio is $\sqrt{2/3} \approx 0.816$.

3. In AutoCAD, an Isometric View (Drawing) is created using: A. Reduced Isometric Scale B. True Scale (1 : 1) C. Double Scale D. Architectural Scale **Answer: B** — Isometric Views use True Scale (1 : 1).

4. Which 3D modeling command sweeps a closed 2D profile around an axis line? A. EXTRUDE B. REVOLVE C. PRESSPULL D. SWEEP **Answer: B** — REVOLVE sweeps profiles around an axis.

5. What receding angle do horizontal isometric axes make with the true horizontal reference

line? A. 15° B. 30° C. 45° D. 60° **Answer: B** — Receding isometric axes make a 30° angle with horizontal.

5.7 G. Unit 5 Quick Revision Checklist

Unit Quick Revision Checklist

- Isometric Axes Angle = 120° between axes, 30° to horizontal.
- Isometric Scale Ratio = $\sqrt{2/3} = \frac{\cos 45^\circ}{\cos 30^\circ} \approx 0.816$.
- Isometric View → True Scale (1:1); Isometric Projection → Reduced Scale (0.816).
- Non-Isometric Lines → Cannot be measured directly; locate endpoints via box coordinates.
- Box Method → Enclose solid in L, W, H rectangular box.
- AutoCAD 3D: EXTRUDE (linear stretch), REVOLVE (axis sweep), PRESSPULL (boundary de

Unit 6: Development of Surfaces & 3D Solid Editing

6.1 A. What You'll Learn

Development of surfaces is the geometric unrolling or unfolding of 3D solid surfaces into a 2D flat plane pattern. In this unit, you will master:

1. Surface development principles and industrial sheet metal applications (ducts, pipes, hoppers, funnels, chimneys).
 2. Governing rules: **Stretch-out Line** (full perimeter) and **True Length (TL)** enforcement.
 3. **Parallel Line Method** for Prisms and Cylinders.
 4. **Radial Line Method** for Pyramids and Cones ($\theta = \frac{r}{L} \times 360^\circ$).
 5. **Triangulation Method** for transition pieces.
 6. Development procedures for regular and truncated solids.
 7. AutoCAD 3D Solid Editing & Boolean operations ('UNION', 'SUBTRACT', 'INTERSECT').
 8. AutoCAD 3D Navigation ('3DORBIT', Visual Styles) and 2D view extraction ('VIEW-BASE').
-

6.2 B. Principles of Surface Development

Definition: Development of Surfaces

The process of unfolding or unrolling the lateral surfaces of a 3D solid object onto a single 2D plane surface to produce a flat pattern. When cut from sheet metal and folded along bend lines, it forms the 3D solid.

6.2.1 Industrial Applications

Sheet metal fabrication of air conditioning ducts, boiler shells, hoppers, funnels, chimneys, automobile body panels, and packaging boxes.

6.2.2 Key Terminology & Governing Rules

- **Stretch-out Line:** The full length or perimeter of the base laid out in a straight line.
- **True Length (TL):** Every line on the development pattern **MUST** represent the true

length** of the corresponding edge on the solid. If an edge is inclined in orthographic views, its true length must be determined before drawing development.

- **Seam Line Selection:** The opening/closing seam line is usually chosen along the **shortest edge** to minimize welding or soldering length.

6.3 C. Methods of Surface Development

Method Name	Applicability (Solids)	Governing Geometric Principle
Parallel Line Method	Prisms and Cylinders	Lateral edges/generators are parallel to each other. Stretch-out line length = Base Perimeter.
Radial Line Method	Pyramids and Cones	Lateral edges slant towards a single apex. Development is a sector of a circle with Apex as center and Slant Edge (TL) as radius.
Triangulation Method	Transition pieces (e.g., square-to-round adapters)	Surface is divided into a series of triangles laid out adjacent to one another.

Table 6.1. Classification of Surface Development Methods.

6.4 D. Development Procedures for Prisms & Pyramids

6.4.1 1. Truncated Prisms (Parallel Line Method)

Prism Stretch-out Line Formula

$$\text{Stretch-out Line Length } (L) = \text{Perimeter of Base} \quad (6.1)$$

$$\text{Square Prism: } L = 4 \times s, \quad \text{Hexagonal Prism: } L = 6 \times s, \quad \text{Cylinder: } L = \pi D$$

Step-by-Step Procedure

1. Draw Top View and Front View of the prism with section cutting plane.
2. Draw horizontal stretch-out line aligned with base of Front View equal to base perimeter.
3. Divide stretch-out line into segments corresponding to base side lengths (s).
4. Erect vertical lines (generators) from division points.
5. Project intersection points of cutting plane from Front View horizontally across to corresponding vertical lines on development.
6. Join projected points with straight lines to form upper boundary of development pattern.

6.4.2 2. Truncated Pyramids & Cones (Radial Line Method)

Cone Development Sector Angle (θ)

For a cone with base radius r and true slant height L :

$$\theta = \left(\frac{r}{L} \right) \times 360^\circ \quad (6.2)$$

Step-by-Step Procedure

1. Draw Top and Front Views. Determine **True Length (TL)** of slant edge. (If slant edge in FV is not parallel to XY , rotate it to find TL). 2. Draw an arc with Apex P as center and radius equal to TL . 3. Step off chord lengths along arc equal to base polygon side length s . 4. Join chord division points to Apex P with straight lines. 5. **For Truncated Pyramids:** Mark cut points on the TL slant edge in Front View. Transfer these distances to radial lines using concentric arcs centered at Apex P .

Worked Example: Cone Development Sector Angle

Problem: Calculate the sector angle (θ) for developing a cone of base diameter 50 mm ($r = 25$ mm) and altitude height 60 mm.

Solution Step 1: Calculate True Slant Height (L)

$$L = \sqrt{r^2 + h^2} = \sqrt{25^2 + 60^2} = \sqrt{625 + 3600} = \sqrt{4225} = 65 \text{ mm}$$

Solution Step 2: Calculate Sector Angle (θ)

$$\theta = \left(\frac{r}{L} \right) \times 360^\circ = \left(\frac{25}{65} \right) \times 360^\circ = 138.46^\circ$$

6.5 E. AutoCAD 3D Solid Editing & Navigation

6.5.1 3D Boolean Operations

COMMAND: UNION

Merges two or more separate 3D solids or regions into a single composite solid.

- **Command:** UNION or UNI
- **Workflow:** Type UNI → Select 3D solids → Press **Enter**.

COMMAND: SUBTRACT

Removes the volume of one 3D solid from another (essential for creating holes, slots, or truncated cuts).

- **Command:** SUBTRACT or SU
- **Workflow:** Type SU → Select solid to KEEP → Press **Enter** → Select solid to REMOVE → Press **Enter**.

COMMAND: INTERSECT

Creates a 3D solid from the overlapping common volume of two or more solids.

- **Command:** INTERSECT or IN

6.5.2 3D Navigation & 2D View Extraction

- **3DORBIT (3D0):** Dynamically rotates 3D view in space (**Shift** + Middle Mouse Drag).
- **Visual Styles (VS):** Controls edge/shading display (2D Wireframe, Conceptual, Realistic, Hidden).
- **VIEWBASE:** Automatically generates linked 2D Orthographic Front, Top, Side, and Isometric views directly from 3D Model Space onto a Layout sheet.

6.6 F. Chapter 6 Practice Problems & MCQs

6.6.1 Practice Questions

Viva Q&A: Question 1 (Basic)

Which development method is used for Prisms and Cylinders? What is the length of the stretch-out line?

Answer: The ****Parallel Line Method****. Length of stretch-out line equals the perimeter of the base ($L = \text{Perimeter}$).

Viva Q&A: Question 2 (Moderate)

Write the formula for the development sector angle (θ) of a cone.

Answer: $\theta = \left(\frac{r}{L}\right) \times 360^\circ$, where r is base radius and L is true slant height.

Viva Q&A: Question 3 (Exam Level)

Calculate the sector angle for developing a cone of base radius 30 mm and true slant height 90 mm.

Answer:

$$\theta = \left(\frac{30}{90}\right) \times 360^\circ = \frac{1}{3} \times 360^\circ = 120^\circ$$

6.6.2 Multiple-Choice Questions (MCQs)

1. Which method of development is used for Pyramids and Cones? A. Parallel Line Method B. Radial Line Method C. Triangulation Method D. Approximate Method **Answer: B** — Pyramids and Cones are developed using Radial Line Method.

2. All lines on a surface development pattern MUST represent: A. Apparent length B. True length C. Foreshortened length D. Isometric length **Answer: B** — Every edge on a development pattern must represent True Length (TL).

3. Which AutoCAD Boolean command removes one 3D solid from another? A. UNION B. SUBTRACT C. INTERSECT D. SLICE **Answer: B** — SUBTRACT removes 3D solid volumes.

4. The stretch-out line length for a cylinder of diameter D is: A. πD B. $2\pi D$ C. πD^2 D. $D/2$ **Answer: A** — Stretch-out line length equals perimeter = πD .

5. Which AutoCAD command automatically generates linked 2D orthographic views from a 3D model? A. EXTRUDE B. PRESSPULL C. VIEWBASE D. 3DORBIT **Answer: C** — VIEWBASE generates 2D views from 3D models.

6.7 G. Unit 6 Quick Revision Checklist

Unit Quick Revision Checklist

- Parallel Line Method → Prisms and Cylinders ($L = \text{Perimeter}$).
- Radial Line Method → Pyramids and Cones ($\theta = \frac{r}{L} \times 360^\circ$).
- Triangulation Method → Transition pieces (square-to-round adapters).
- True Length Rule → All edges on development MUST represent True Length TL .
- Seam Line → Select shortest edge to minimize welding/soldering.
- AutoCAD Booleans: UNION (combine), SUBTRACT (cut hole), INTERSECT (common volume).
- VIEWBASE → Auto-generates 2D views from 3D Model Space onto Layout Tab.

Chapter 7: MEC136 Master Revision Sheet

7.1 Master Formula & Equation Reference

Formula Name	Mathematical Equation	Operational Variables & Notes
Representative Fraction	$R.F. = \frac{\text{Drawing Length}}{\text{Actual Length}}$	Units of numerator and denominator must match before reducing fraction.
Length of Scale	$L.O.S. = R.F. \times \text{Max Length}$	Used to calculate physical line length for scale on paper.
Lettering Spacing	Spacing = $\frac{1}{5}h$, Line = $0.5h$ to $1h$	Single-stroke vertical Gothic, 7 : 4 height-to-width ratio.
Isometric Scale Factor	$\frac{\text{Iso Length}}{\text{True Length}} = \frac{\cos 45^\circ}{\cos 30^\circ} = \sqrt{\frac{2}{3}} \approx 0.816$	Isometric Length = $0.816 \times$ True Length.
Front View Apparent Length	$a'b' = TL \cos \phi$	Where ϕ is inclination of line to VP.
Top View Apparent Length	$ab = TL \cos \theta$	Where θ is inclination of line to HP.
Cylinder Stretch-out Length	$L = \pi D$	Parallel Line Method base perimeter for cylinder.
Cone Sector Angle	$\theta = \left(\frac{r}{L}\right) \times 360^\circ$	Where r = base radius, L = true slant height.

Table 7.1. Master Formula Reference Sheet.

7.2 Projection Rules & Quadrant Quick Reference

7.3 AutoCAD Command Cheat Sheet

System / Quadrant	View Layout Rules	Key Characteristics
First Angle (ISO)	FV above XY , TV below FV, LSV to right of FV	Object between Observer and Plane. Standard in India/Europe.
Third Angle (ANSI)	FV below XY , TV above FV, RSV to right of FV	Plane between Observer and Object. Standard in USA/Canada.
1st Quadrant Point	a' above XY , a below XY	Point above HP, in front of VP.
2nd Quadrant Point	a' above XY , a above XY	Views overlap above reference line XY .
3rd Quadrant Point	a' below XY , a above XY	Point below HP, behind VP.
4th Quadrant Point	a' below XY , a below XY	Views overlap below reference line XY .

Table 7.2. Projection System and Point Location Rules.

Command	Shortcut	Primary Purpose	Key Option / Trap
UNITS	UN	Set precision and unit types	Set Insertion Scale to Millimeters.
LIMITS	—	Set drawing boundaries	Must follow with ZOOM ALL ($Z \rightarrow A$).
OSNAP	F3	Object Snap toggles	Enable Endpoint, Midpoint, Center.
ORTHO	F8	Restrict motion to horizontal/vertical	Essential for straight projection lines.
LINE	L	Create straight line segments	Polar syntax: @length<angle.
PLINE	PL	Create connected 2D polyline	Forms a single unified object.
TRIM	TR	Cut objects to boundary edge	Select cutting edge $\rightarrow$ click object.
FILLET	F	Round sharp corner edges	Set radius first: F $\rightarrow$ R $\rightarrow$ Value.
CHAMFER	CHA	Bevel sharp corner edges	Set distance first: CHA $\rightarrow$ D $\rightarrow$ Values.
STRETCH	S	Elongate geometry	MUST use Green Crossing Window.
HATCH	H	Fill section with pattern	Pattern ANSI31 at 45°, Pick Points.
EXTRUDE	EXT	Convert 2D profile to 3D solid	Input must be closed polyline/region.
REVOLVE	REV	Sweep 2D profile around axis	Requires closed profile and axis line.
PRESSPULL	—	Dynamic 3D volume pull	Auto-detects enclosed coplanar area.
SUBTRACT	SU	Cut 3D volume	Select KEEP solid $\rightarrow$ Select REMOVE solid.
VIEWBASE	—	Generate 2D views from 3D model	Auto-links Model Space to Layout Tab.

Table 7.3. AutoCAD Command Cheat Sheet.

7.4 Common Exam Traps & Pitfalls

1. **IC 7402 / Sectioning Misconceptions:** In Sectional Views, thin webs/ribs cut longitudinally are ****NOT**** hatched. Fasteners (bolts, nuts, screws) and solid shafts cut longitudinally are ****NOT**** hatched.
2. ****Isometric View vs. Projection:**** Isometric ****View**** uses True Scale (1 : 1). Isometric ****Projection**** uses reduced scale factor (0.816). Do not confuse the two when reading exam instructions!
3. ****Line Projection Projector Alignment:**** In the Rotating Line Method, Front View end point b' and Top View end point b ****MUST** lie on the same vertical projector line^{**}. If they do not align, the construction is incorrect.
4. ****AutoCAD Stretch Command Error:**** Selecting objects with a single click or a Blue Window fails for **STRETCH**. You ****MUST**** use a Green Crossing Window (dragged right-to-left).

Chapter 8: Integrated Synthesis & Practical Problems

8.1 Overview

This chapter contains 15 comprehensive, multi-concept synthesis problems combining manual drafting theory, geometric derivations, line projections, sectioning rules, isometric views, surface developments, and AutoCAD drafting workflows.

8.2 Integrated Synthesis Problem Sets

8.2.1 Problem 1: Scale Construction & Line Projection

Problem Statement: A line AB is drawn on a map constructed to a scale of $R.F. = 1 : 50$. The front view of line AB measures 60 mm and is inclined at 30° to the XY reference line. If the line is parallel to the VP and its end A is 20 mm above the HP, determine: (a) The actual physical length of line AB , and (b) The inclination of line AB with the HP (θ).

Detailed Solution:

1. **Analyze Orientation:** Since line AB is parallel to the VP, its Front View shows the ****True Length (TL)**** on the drawing sheet.
2. ****Drawing Sheet Length:**** $TL_{\text{drawing}} = 60 \text{ mm}$.
3. ****Calculate Actual Physical Length:****

$$R.F. = \frac{\text{Length on drawing}}{\text{Actual length}} \implies \frac{1}{50} = \frac{60 \text{ mm}}{\text{Actual Length}}$$

$$\text{Actual Physical Length} = 60 \text{ mm} \times 50 = 3000 \text{ mm} = \mathbf{3.0 \text{ m}}$$

4. ****Inclination with HP (θ):**** Since the Front View is inclined at 30° to XY and line is parallel to VP, the true inclination with HP is $\theta = \mathbf{30^\circ}$.
-

8.2.2 Problem 2: Oblique Line Projection & Traces

Problem Statement: A line CD has its end C 15 mm above HP and 20 mm in front of VP. The line is 80 mm long and inclined at 45° to HP and 30° to VP. Draw the orthographic projections of line CD using the Rotating Line Method, find its apparent angles α and β , and locate its Horizontal Trace (HT) and Vertical Trace (VT).

Detailed Solution:

1. **Given Data:** $TL = 80 \text{ mm}$, $c' = 15 \text{ mm}$ above XY , $c = 20 \text{ mm}$ below XY , $\theta = 45^\circ$, $\phi = 30^\circ$.

2. Calculate Apparent Projections:

Top View Apparent Length (cd_1) = $TL \cos \theta = 80 \cos 45^\circ = 80 \times 0.7071 = \mathbf{56.57 \text{ mm}}$

Front View Apparent Length ($c'd'_2$) = $TL \cos \phi = 80 \cos 30^\circ = 80 \times 0.8660 = \mathbf{69.28 \text{ mm}}$

Height Difference (Δh) = $TL \sin \theta = 80 \sin 45^\circ = \mathbf{56.57 \text{ mm}}$

Depth Difference (Δd) = $TL \sin \phi = 80 \sin 30^\circ = \mathbf{40.00 \text{ mm}}$

- Construction & Angles:** Following Rotating Line construction yields apparent angles $\alpha \approx 54.7^\circ$ in FV and $\beta \approx 45.0^\circ$ in TV.
- Trace Location:** Extend $c'd'$ to meet XY at h ; project vertically down to meet extended cd to locate HT. Extend cd to meet XY at v ; project vertically up to meet extended $c'd'$ to locate VT.

8.2.3 Problem 3: Orthographic Projection & Miter Line Method

Problem Statement: A stepped block consists of a base rectangle 60 mm $\times$ 40 mm $\times$ 20 mm high, topped with a central cylinder of diameter 30 mm and height 25 mm. Describe the step-by-step procedure to construct First Angle Front View, Top View, and Right Side View using the 45° Miter Line Method in AutoCAD.

Detailed Solution:

- Front View (FV):** Draw base rectangle width 60 mm $\times$ height 20 mm. On top edge, draw cylinder rectangle width 30 mm $\times$ height 25 mm centered horizontally. Total height = 45 mm.
- Top View (TV):** Positioned directly below FV. Draw base rectangle 60 mm $\times$ 40 mm. Draw a circle diameter 30 mm at center (30, 20).
- Miter Line Setup:** Draw reference lines XY and vertical Y_1Y_1 . From intersection point, draw Miter Line at 45° into top-right quadrant.
- Side View Transfer:** Project horizontal lines from TV rightwards to meet Miter Line. Project vertically down into Right Side View quadrant (drawn to left of FV in 1st Angle). Intersect with horizontal projection lines from FV to complete Right Side View.

8.2.4 Problem 4: Offset Sectional View & Hatching Exceptions

Problem Statement: An offset sectional view is taken through a cast iron casing containing a central 20 mm hole, a parallel 15 mm hole offset by 35 mm, a longitudinal reinforcing web of thickness 10 mm, and an M12 assembly bolt. Explain the hatching rules for each feature.

Detailed Solution:

- Offset Cutting Plane:** The cutting plane line steps horizontally to pass through both offset holes. The bends/steps are ****omitted**** in the final sectional view.
- Section Hatching (ANSI31):** Solid cast iron walls cut by plane are hatched with thin continuous lines at 45°.
- Holes:** The 20 mm and 15 mm void holes are ****NOT**** hatched.
- Reinforcing Web Exception:** The 10 mm web cut longitudinally is ****NOT**** hatched.
- Assembly Bolt Exception:** The M12 steel bolt cut longitudinally is ****NOT**** hatched.

8.2.5 Problem 5: Isometric View vs. Projection of Composite Solid

Problem Statement: A regular square pyramid of base side 40 mm and height 50 mm rests coaxially on top of a square prism of base side 60 mm and height 30 mm. Calculate: (a) The true dimensions for drawing an **Isometric View**, and (b) The reduced dimensions for drawing an **Isometric Projection**.

Detailed Solution:

- Part (a): Isometric View (True Scale 1 : 1)** - Prism Base = 60 mm, Prism Height = 30 mm. - Pyramid Base = 40 mm, Pyramid Height = 50 mm. - Dimensions are drawn **100% at full actual size**.
- Part (b): Isometric Projection (Isometric Scale 0.816)** - Reduced Prism Base = $0.816 \times 60 = 48.96$ mm. - Reduced Prism Height = $0.816 \times 30 = 24.48$ mm. - Reduced Pyramid Base = $0.816 \times 40 = 32.64$ mm. - Reduced Pyramid Height = $0.816 \times 50 = 40.80$ mm.

8.2.6 Problem 6: Truncated Cone Surface Development

Problem Statement: A cone of base diameter 60 mm ($r = 30$ mm) and height 70 mm is cut by a section plane inclined at 45° to HP, passing through the axis at a height of 30 mm from base. Calculate sector angle θ and outline the Radial Line Method development steps.

Detailed Solution:

- Calculate True Slant Height (L):**

$$L = \sqrt{r^2 + h^2} = \sqrt{30^2 + 70^2} = \sqrt{900 + 4900} = \sqrt{5800} = 76.16 \text{ mm}$$

- Calculate Sector Angle (θ):**

$$\theta = \left(\frac{r}{L} \right) \times 360^\circ = \left(\frac{30}{76.16} \right) \times 360^\circ = 141.8^\circ$$

- Development Construction:** Draw sector of radius 76.16 mm and angle 141.8° . Divide base arc into 12 equal parts. Transfer cut points from TL slant edge in FV using concentric arcs centered at Apex to cut corresponding radial lines.

8.2.7 Problem 7: AutoCAD 3D Solid Modeling & Boolean Editing

Problem Statement: Detail the AutoCAD command sequence to model a hollow rectangular block of size $100 \times 80 \times 50$ mm with a central cylindrical hole of diameter 40 mm passing through it completely.

Detailed Solution:

- Switch workspace to **3D Modeling**. Set view to **SE Isometric**.
- Create outer block: Type **BOX** $\rightarrow 0,0,0 \rightarrow$ Length = 100, Width = 80, Height = 50.
- Create inner cylinder: Type **CYLINDER** $\rightarrow$ Center point = 50, 40, 0 $\rightarrow$ Radius = 20 (Diameter 40) $\rightarrow$ Height = 50.
- Subtract cylinder to create hole: Type **SUBTRACT** (or **SU**) $\rightarrow$ Select **BOX** (solid to keep) $\rightarrow$ Press **Enter** $\rightarrow$ Select **CYLINDER** (solid to remove) $\rightarrow$ Press **Enter**.
- Change Visual Style to **Conceptual** (**VS** $\rightarrow$ **C**) to verify hole.

Chapter 9: Complete Full-Length Mock Examinations

9.1 Overview

This chapter provides two complete, full-length university examination papers modeled on the official LPU examination pattern. Each paper contains Section A (Objective/MCQs), Section B (Short Answer & Derivation Problems), and Section C (Long Construction & AutoCAD Problems) with complete marking rubrics, answer keys, and step-by-step solutions.

9.2 MEC136 MOCK EXAMINATION PAPER 1

Time Allowed: 3 Hours • **Maximum Marks:** 100 Marks

9.2.1 SECTION A: Objective & Multiple-Choice Questions (20 Marks)

Answer all 10 questions. Each question carries 2 Marks.

1. What is the *R.F.* of a scale if 1 cm represents 5 m? (A) 1 : 5 (B) 1 : 50 (C) 1 : 500
(D) 1 : 5000
 2. In First Angle Projection, the Top View is placed: (A) Above Front View (B) Below Front View (C) To the left of Front View (D) On *XY* line
 3. The isometric length of an edge whose true length is 100 mm is: (A) 81.6 mm (B) 86.6 mm (C) 70.7 mm (D) 100 mm
 4. Which line type is used for center lines and axes of symmetry? (A) Continuous Thick
(B) Dashed Medium (C) Chain Thin (D) Continuous Thin
 5. What is the sector angle θ for developing a cone of base radius 20 mm and slant height 80 mm? (A) 45° (B) 90° (C) 120° (D) 180°
 6. In AutoCAD, which command sets the invisible drawing area boundaries? (A) UNITS
(B) LIMITS (C) ZOOM (D) GRID
 7. Which solid component is NOT hatched when cut longitudinally? (A) Hollow Pipe (B) Thin Reinforcing Web (C) Casing Wall (D) Flange Rim
 8. Which AutoCAD modifier command requires a green Crossing Window selection? (A) TRIM (B) EXTEND (C) STRETCH (D) MIRROR
 9. The angle between any two principal isometric axes is: (A) 30° (B) 60° (C) 90°
(D) 120°
 10. Which development method is used for Pyramids and Cones? (A) Parallel Line Method
(B) Radial Line Method (C) Triangulation Method (D) Box Method
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9.2.2 SECTION B: Short Answer & Analytical Problems (30 Marks)

Answer any 3 questions. Each question carries 10 Marks.

11. Derive the mathematical reduction factor (0.816) for an Isometric Scale from first principles of trigonometry. 12. A line AB , 70 mm long, has its end A 15 mm above HP and 20 mm in front of VP. The line is parallel to VP and inclined at 45° to HP. Draw its Front View and Top View, and calculate its apparent top view length. 13. Differentiate between Full Section, Half Section, and Offset Section with neat sketches. State two hatching exception rules. 14. Explain the function of AutoCAD drawing aids: OSNAP (F3), ORTHO (F8), LTSCALE, and PRESSPULL.

9.2.3 SECTION C: Long Construction & CAD Workflow Problems (50 Marks)

Answer any 2 questions. Each question carries 25 Marks.

15. ****Line Projection (25 Marks):**** A line PQ , 85 mm long, has its end P 20 mm above HP and 15 mm in front of VP. The front view of the line measures 65 mm and the top view measures 70 mm. Draw the orthographic projections of line PQ using the Rotating Line Method. Determine: (a) True inclination with HP (θ), (b) True inclination with VP (ϕ), and (c) Locate its Horizontal Trace (HT) and Vertical Trace (VT).

16. ****Surface Development (25 Marks):**** A square prism of base side 40 mm and height 65 mm rests vertically on its base on HP with one base edge inclined at 30° to VP. It is cut by a section plane inclined at 45° to HP, perpendicular to VP, and bisecting the vertical axis. Draw: (a) Sectional Front View, (b) Top View, and (c) Full development of lateral surfaces using the Parallel Line Method.

17. ****AutoCAD 3D & 2D Extraction (25 Marks):**** Detail the complete step-by-step AutoCAD command sequence and workflow to model a 3D composite solid consisting of a square pyramid (base 40 mm, height 50 mm) resting coaxially on top of a cylinder (diameter 60 mm, height 40 mm). Explain how to extract 2D orthographic views onto a Layout sheet using VIEWBASE.

9.3 ANSWER KEY & DETAILED SOLUTIONS — PAPER 1

9.3.1 Section A Answer Key

1. ****C (1 : 500)**** — $R.F. = 1 \text{ cm}/500 \text{ cm} = 1/500$. 2. ****B (Below Front View)**** — First angle projection layout. 3. ****A (81.6 mm)**** — Isometric length = $0.816 \times 100 = 81.6 \text{ mm}$. 4. ****C (Chain Thin)**** — Long-short-long chain line for center lines. 5. ****B (90°)**** — $\theta = (r/L) \times 360^\circ = (20/80) \times 360^\circ = 90^\circ$. 6. ****B (LIMITS)**** — Sets paper boundaries. 7. ****B (Thin Reinforcing Web)**** — Longitudinal web cuts are left unhatched. 8. ****C (STRETCH)**** — Requires green Crossing Window selection. 9. ****D (120°)**** — Isometric axes meet at 120° . 10. ****B (Radial Line Method)**** — Used for pyramids and cones.

9.3.2 Section B & C Key Solutions

Solution to Question 11 (Isometric Scale Factor Derivation): Consider a square top face of a cube tilted at 45° to the true horizontal line and 30° to the isometric axis.

$$\text{Isometric Length} = BD \cos 45^\circ, \quad \text{True Length} = BD \cos 30^\circ$$

$$\frac{\text{Isometric Length}}{\text{True Length}} = \frac{\cos 45^\circ}{\cos 30^\circ} = \frac{1/\sqrt{2}}{\sqrt{3}/2} = \sqrt{\frac{2}{3}} \approx \mathbf{0.8165}$$

Solution to Question 12 (Line Parallel to VP): Front View length = 70 mm at 45° to XY . Top View is horizontal length = $70 \cos 45^\circ = 70 \times 0.7071 = \mathbf{49.5 \text{ mm}}$.

Solution to Question 15 (Line Inclined to Both Planes): 1. $TL = 85$ mm, $a'b' = 65$ mm, $ab = 70$ mm. 2. True inclination with HP: $\cos \theta = \frac{\text{Top View Length}}{TL} = \frac{70}{85} = 0.8235 \implies \theta = \mathbf{34.56^\circ}$. 3. True inclination with VP: $\cos \phi = \frac{\text{Front View Length}}{TL} = \frac{65}{85} = 0.7647 \implies \phi = \mathbf{40.12^\circ}$.